C U R R I C U L U M   V I T A E

Rahul Rawat

Working Student, AI Training and Enablement

PERSONAL DETAILS

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Portfolio	rah-portfolio.pages.dev
Date of birth	20 February 2002
Nationality	Indian, student visa with valid work permit
Availability	Werkstudent 20 hours per week immediately, full time from April 2027

PROFILE

Data Science and Analytics Master student at SRH Heidelberg based in Mannheim, with hands on generative AI system building, EU AI Act aware model design, and clear communication of complex technical topics to non technical audiences. I have shipped a modular Retrieval Augmented Generation system with a custom decision making router on Llama **3.1 8b** via Groq and LangChain, a fairness by design credit scoring system documented against EU AI Act and GDPR with a plain language LLM generated explanation for finance managers, and a Random Forest driven climate risk study translated into visual reports for non technical stakeholders. Fluent in English and confident in PowerPoint and design tools, I am the right fit for building AI literacy content and rolling out training at Knauf Digital.

PROFESSIONAL EXPERIENCE

Oct 2025 to Mar 2026
Heidelberg

eRay GmbH

Data Scientist

- Built an end to end recursive time series pipeline forecasting chlorophyll a, turbidity, pH, and dissolved oxygen for a German lake, delivered as a six month collaboration with SRH University Heidelberg.
- Benchmarked six models head to head, Ridge, Gradient Boosting, LightGBM, XGBoost, CatBoost, and Prophet, and used CatBoost multi quantile regression to produce asymmetric **80 percent** prediction intervals for decision support under uncertainty.
- Enforced strict anti leakage rules across the pipeline, surfacing the honest finding that pH and dissolved oxygen are physically predictable while chlorophyll a and turbidity are not without live optical sensors.
- Reconstructed missing winter readings with MICE imputation and engineered a synthetic winter decay forecast canvas so tree based models stopped flatlining during recursive prediction.
- Wrapped the whole pipeline in an orchestrator with gate checks and velocity and ecological bounds that halts on failed imputation rather than letting bad data cascade downstream.

Technologies: *Python, CatBoost, Prophet, scikit learn, MICE*

PERSONAL PROJECTS

2025

Hybrid RAG Orchestrator with Agentic Routing

- Delivered a modular Retrieval Augmented Generation system that classifies user intent into three execution paths, local knowledge retrieval, external web search, or direct conversational logic, measured by clean end to end responses across all three routes, using a custom decision making router on top of Llama **3.1 8b** via Groq and LangChain orchestration.
- Built a persistent semantic memory over PDF and vector data, measured by consistent retrieval on cross session queries, using ChromaDB with local persistence and HuggingFace MiniLM L6 v2 embeddings.
- Kept multi turn context intact for the LLM, measured by coherent follow up answers rather than isolated single shot responses, by adding a stateful MemoryAgent directly into the inference pipeline.
- Shipped the whole system as a Streamlit interface with end to end ownership, measured by a working deployed prototype, using Groq inference and a modular routing layer that can be extended per new intent.

Technologies: *Python, LangChain, Llama 3.1 8b via Groq, ChromaDB, HuggingFace MiniLM L6 v2, Streamlit*

2025

CreditIQ: Fairness by Design Credit Scoring

- Raised the model's Disparate Impact ratio from a failing **0.79** to a compliant **0.88**, measured against the EU AI Act and AGG **80 percent** fairness threshold, using AIF360 mitigation and threshold calibration on a real credit dataset.
- Diagnosed a hidden intersectional bias where younger women were still penalised even after single axis age bias was fixed, measured through SHAP driven subgroup analysis, then designed a four way age by gender threshold matrix to correct it without over correcting into reverse discrimination.
- Cut the false negative rate from **44 percent** to **16.7 percent** while holding accuracy at **75 percent**, measured on the held out test split, by documenting the fairness accuracy trade off as a deliberate and regulator defensible decision rather than a hidden compromise.
- Shipped a Streamlit decision support tool that gives finance managers a recommendation plus a plain language LLM generated explanation, measured against GDPR Article 22 and EU AI Act Article 14 human in the loop requirements, and backed the pipeline with unit tests at **100 percent** branch coverage and a full regulatory write up.

Technologies: *Python, scikit learn, AIF360, SHAP, Streamlit*

2025

Economic Impact Analysis of Global Climate Events

- Executed an end to end data science project transforming raw global event datasets into structured insights for resource allocation and risk assessment, measured by a single reproducible pipeline from raw CSV to a stakeholder report.
- Performed advanced data preparation and cleansing over outliers, imputation, and normalisation, measured by stable model performance before and after feature scaling, to build a high quality data foundation.
- Developed Random Forest models to analyse correlations between event duration and financial impact, measured by feature importance rankings and residual analysis, extracting clear business relevant insights on where economic risk concentrates.
- Communicated complex statistical findings to non technical stakeholders, measured by comprehensive visual reports and calibrated confidence statements that could survive a management review.

Technologies: *Python with Pandas and scikit learn, Random Forest, statistical modelling, Matplotlib and Seaborn*

EDUCATION

Apr 2025 to Present
Heidelberg

M.Sc. Data Science and Analytics

SRH University of Applied Sciences Heidelberg, GPA 1.9

2019 to 2023
Greater Noida, India

Bachelor of Technology in Computer Science

GL Bajaj Institute of Technology and Management, CGPA 7.3 of 10

RESEARCH AND THESIS

2022 to 2023
Greater Noida, India

Bachelor Thesis, Diabetes Prediction Using Machine Learning

GL Bajaj Institute of Technology and Management, IEEE style paper

- Built a full end to end machine learning pipeline comparing six classifiers on a clinical dataset of **768 patients**, using **10 fold** cross validation and per model confusion matrices to make the model comparison auditable.
- Caught biologically impossible zero values that the original authors had overlooked, then applied IQR based outlier removal and proper imputation to restore data integrity before any model fit.
- Chose ROC AUC over accuracy as the headline metric for a **65 to 35** imbalanced dataset, avoiding a misleadingly rosy accuracy figure that would have hidden real error patterns in the minority class.
- Wrote up findings as an IEEE style paper including an honest section on limitations and what to do differently in a follow up study, publishable in substance.

Technologies: *Python, scikit learn, Pandas, Seaborn*

CERTIFICATIONS

- NVIDIA: Building LLM Applications With Prompt Engineering, issued November 2025.
- AWS Academy Graduate: AWS Academy Cloud Foundations, issued July 2025.
- Google Data Analytics: Foundations, Data, Data, Everywhere, issued April 2025 on Coursera.

ACHIEVEMENTS

- USAII Global AI Hackathon 2026: Finalist at Graduate Level, awarded by the United States Artificial Intelligence Institute for innovation, technical creativity, and applied AI on real world challenges.

LANGUAGES

English	Fluent, written and spoken
German	B1 in progress toward B2
Hindi	Native

Mannheim, 23 July 2026
Rahul Rawat